Requirement Analysis

Solution Requirements (Functional & Non-Functional)

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,” focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether it's through targeted promotional campaigns, adjustments in product formulations, or personalized recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

Functional Requirements

FR #	Requirement	Description
FR-01	Data Connectivity	System shall connect to the cleaned cosmetics product dataset (CSV/Excel, 1,472 rows)
FR-02	Data Cleaning	System shall standardize brand-name casing, replace scrape-artifact ingredient text with proper missing values, and treat a Rank of 0 as unrated
FR-03	Data Reshaping	System shall pivot the five skin-type indicator columns into two long-format fields, Skin_Type and Suitable, for dynamic filtering
FR-04	Category Filter	System shall allow users to filter all visualizations by product category (Label)
FR-05	Brand Filter	System shall provide a searchable, checkbox-based Brand filter applied across the dashboard
FR-06	Avg Price by Category Chart	System shall display a bar chart of average price per product category
FR-07	Price vs Rating Chart	System shall display a scatter plot of average price against average rating, colour-coded by category
FR-08	Top Brands by Category Chart	System shall display a bar chart ranking the top 10 brands by average rating

FR #	Requirement	Description
FR-09	Skin Type Coverage Chart	System shall display a heatmap of average suitability by category and skin type
FR-10	Cross-Filter Actions	System shall enable click-to-filter behaviour so selecting a mark on any chart filters the other three
FR-11	Dashboard Assembly	System shall combine all four charts into a single fixed-size (1366×768) interactive dashboard
FR-12	Dashboard Title	System shall display a descriptive dashboard title (“Cosmetic Insights: Trends & Consumer Behavior”)
FR-13	Tooltip on Hover	All charts shall display relevant field values (brand, price, rating, category) on hover
FR-14	Tableau Public Publishing	System shall be published to Tableau Public for browser-based access without login

Non-Functional Requirements

NFR #	Category	Requirement
NFR-01	Usability	Dashboard UI shall be navigable by non-technical users (consumers, retail staff) without training
NFR-02	Performance	All filters and cross-chart actions shall respond and re-render within 2 seconds on the full dataset (7,360 rows post-pivot)
NFR-03	Reliability	Published Tableau Public workbook shall be available and accessible via its public URL at all times
NFR-04	Scalability	Solution architecture shall support addition of new products or brands without redesigning dashboards
NFR-05	Availability	Dashboard shall be accessible via standard web browsers (Chrome, Edge, Safari) without plugins or login
NFR-06	Security	No personally identifiable information (PII) is stored or displayed; the dataset is fully public product data
NFR-07	Cost	The entire pipeline shall run on free tooling — Python/Pandas for cleaning and Tableau Public for visualization
NFR-08	Compatibility	Solution shall render correctly on desktop (1366×768 or higher) and in Tableau Public's mobile view